

Career Roadmap: AI / Machine Learning Engineer

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Phase 1: Mathematics & Python Foundations

Linear Algebra, Multivariable Calculus, Probability, NumPy, and Scikit-Learn.

Phase 2: Supervised & Unsupervised Machine Learning

Regression, Classification, Decision Trees, Random Forests, XGBoost, K-Means clustering, and Model Evaluation metrics.

Phase 3: Deep Learning & Neural Networks

PyTorch / TensorFlow, Convolutional Neural Networks (CNN), Recurrent Networks (RNN/LSTM), and Transformers.

Phase 4: Generative AI & RAG Engineering

LLM fine-tuning, Prompt Engineering, Vector Databases (FAISS), Embedding Models, and LangChain/LlamaIndex paradigms.